

COSC 594/650

KKT Conditions

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Lagrangian duality

$$\begin{array}{ll}\underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & x \in X\end{array}$$

■ LAGRANGIAN AND DUAL FUNCTION

■ WEAK AND STRONG DUALITY

Problem	Weak duality	Strong duality
General	$d^* \leq p^*$	may fail
Convex + Slater	$d^* \leq p^*$	$d^* = p^*$

QUESTION: Is the dual problem easier to solve?

Gradient projection on the dual

Primal problem: minimize $\frac{1}{2}\|x\|^2$ subject to $Ax \leq b$

Dual problem: maximize $-\frac{1}{2}\lambda^\top AA^\top \lambda - b^\top \lambda$ subject to $\lambda \geq 0$

■ PROJECTION

- ★ primal: onto the polyhedron $\{x \mid Ax \leq b\}$ – a quadratic program
- ★ dual: onto the orthant $\{\lambda \mid \lambda \geq 0\}$ – $[\lambda]_+ = \max\{\lambda, 0\}$

■ DUAL GRADIENT PROJECTION

$$\lambda_{k+1} = \left[\lambda_k - \alpha (AA^\top \lambda_k + b) \right]_+$$

- ★ $x_k = -A^\top \lambda_k$ – primal iterate

Strong duality: $x^\star = -A^\top \lambda^\star$ minimizes $L(x, \lambda^\star)$

Stationarity

Strong duality: $f_0(x^*) = g(\lambda^*)$, x^* and λ^* optimal

$$\begin{aligned} f_0(x^*) &= \inf_{x \in X} \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) \right) \\ &\leq f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) \\ &\leq f_0(x^*) \end{aligned}$$

$$\Rightarrow \left\langle \nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*), x - x^* \right\rangle \geq 0, \quad x \in X$$

Complementary slackness

Strong duality: $f_0(x^*) = g(\lambda^*)$, x^* and λ^* optimal

$$\begin{aligned}f_0(x^*) &= \inf_{x \in X} \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) \right) \\&\leq f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) \\&\leq f_0(x^*) \\ \implies \lambda_i^* f_i(x^*) &= 0 \text{ for } i = 1, \dots, m\end{aligned}$$

Constraint	$f_i(x^*)$	Multiplier
inactive	< 0	$\lambda_i^* = 0$
active	$= 0$	$\lambda_i^* \geq 0$

Karush–Kuhn–Tucker conditions

Differentiable functions: $f_0, f_1, \dots, f_m$ over $X = \mathbb{R}^n$

Zero duality gap: (x^*, λ^*) primal and dual optimal

■ KKT CONDITIONS

- ★ $f_i(x^*) \leq 0, \quad \forall i$ – primal feasibility
- ★ $\lambda_i^* \geq 0, \quad \forall i$ – dual feasibility
- ★ $\nabla f_0(x^*) + \sum_i \lambda_i^* \nabla f_i(x^*) = 0$ – stationarity
- ★ $\lambda_i^* f_i(x^*) = 0, \quad \forall i$ – complementary slackness

CONVEXITY:

KKT conditions $\iff$ Optimality with zero duality gap

Worked example

Nonnegative trust weights: $f_j(\theta) = -\theta_j \leq 0$ – affine constraints

$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & f(\theta) = \|X\theta - y\|^2 \\ \text{subject to} & \theta \geq 0 \end{array}$$

■ KKT CONDITIONS

$$\nabla f(\theta^*) = \lambda^* \geq 0, \quad \lambda_j^* \theta_j^* = 0, \quad \theta^* \geq 0$$

■ OPTIMALITY

$$\begin{aligned} \theta_j^* > 0 &\implies \frac{\partial f(\theta^*)}{\partial \theta_j} = 0 \\ \theta_j^* = 0 &\implies \frac{\partial f(\theta^*)}{\partial \theta_j} \geq 0 \end{aligned}$$

KKT recovers the optimality condition over the nonnegative orthant

Lagrangian duality and KKT conditions

$$\begin{array}{ll}\underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & x \in X\end{array}$$

■ LAGRANGIAN AND DUAL FUNCTION

■ WEAK AND STRONG DUALITY

■ KARUSH–KUHN–TUCKER CONDITIONS

Problem	Weak duality	Strong duality	KKT conditions
General	$d^* \leq p^*$	may fail	necessary if $d^* = p^*$
Convex + Slater	$d^* \leq p^*$	$d^* = p^*$	necessary & sufficient

Questions?

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